

By Theorem 12(a), a transformation T is linear and onto ~~from~~ $\mathbb{R}^n$ if and only if the columns of the standard matrix A span $\mathbb{R}^m$.

By Theorem 4, the columns of the standard matrix A span $\mathbb{R}^m$ if and only if there is a pivot on every row of A .

By defn, a pivot corresponds to a leading 1 of a row in rref.

25) The transformation in Exercise 17.

Solution:

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow[\text{Row 1 and 2}]{\text{Interchange}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow[\text{Row 2 and 4}]{\text{interchange}}$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Row 2 and 3}]{\text{interchange}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{-Row 3}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[\text{-Row 2}]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Verification of whether the final matrix is in rref

1. Row 1 to 3 are non-zero rows. Row 4 is a zero row. $\therefore$ The condition that all the non-zero rows must be above the zero rows is True.

~~Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0~~
~~Row 2 Column 2 = Row 4 Column 2 = 0~~
~~Row 4 Column 3 = 0~~

2. Row 1 Leading entry = 1 in Column 1
 Row 2 Leading entry = 1 in Column 2
 Row 3 Leading entry = 1 in Column 3

$\therefore$ The condition that for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True

3. Row 2 Column 1 = Row 3 Column 1 = Row 4 Column 1 = 0
 Row 3 Column 2 = Row 4 Column 2 = 0
 Row 4 Column 3 = 0